⚠ ALL DOCUMENTATION MUST BE APPROVED BY WEDNESDAY, 25 MARCH 2026

CONSULTING PROPOSAL

Applied AI Consulting & Development

South African Sugar Association

BUDGET LINE: PROFESSIONAL FEES

PREPARED FOR	SESSIONS	INVESTMENT
SASA	**6 × 7 hours**	**R24,150 incl. VAT**
DATE	REFERENCE	PREPARED BY
23 March 2026	**PROP-SASA-2026-006**	**Richard Shewry**

01 — OVERVIEW

Executive Summary

This proposal outlines a structured six-session Applied AI programme for the South African Sugar Association. Over approximately eleven weeks, your teams will move from AI awareness to operational capability — producing real outputs in each session, not just attending presentations.

The programme is built around SASA's actual data systems: R7106, LIMS, Autolab, and mill reporting workflows. Every session is grounded in real problems, real documents, and real outputs. By the final session, teams will have built and shipped at least one working tool.

6 SESSIONS	**~11** WEEKS	**26** 15 IN-PERSON + 11 ONLINE
6 DELIVERABLES	**R0** ONGOING TOOLS COST	**R21k** EXCL. VAT

02 — METHOD

Approach

This is not a training course. It is a consulting engagement where learning happens through doing. The principles below govern how every session is designed and delivered.

- **Grounded in your data.** Every exercise uses SASA's real systems and real documents — not synthetic examples or generic datasets.
- **Output in every session.** Each session ends with a tangible deliverable that SASA staff can use from the following Monday.
- **Managers direct, AI writes.** The programme builds prompting and oversight skills, not dependency. Your team learns to use AI as a junior analyst, not a black box.
- **Zero-cost tool stack.** Where possible, all tools are open-source and self-hosted (Python, n8n, Ollama, Open WebUI). No per-seat licensing beyond Claude Pro.
- **Build, don't present.** The final session is a live build day. Teams ship working tools, not slide decks.
- **Governance from day one.** Data classification, acceptable use, and prompt ownership are introduced in Session 1 and formalised in Session 6.
- **Senior consultant throughout.** Richard leads all sessions. Advanced participants co-facilitate in Session 6 — building internal capability, not dependency on the consultant.

03 — PROGRAMME

Programme Overview — Six Sessions, One Direction

The six sessions run fortnightly over eleven weeks, building from foundations to live deployment. Each session is seven hours, split across morning and afternoon.

REF	SESSION TITLE	TIMING
S1	Foundations Audit, Strategic Framing & Prompt Engineering	Week 1
S2	Excel, Reporting & AI-Generated Workflows	Week 3
S3	Working with Large Datasets, RAG & Context Management	Week 5
S4	Python & Automation Pipelines	Week 7
S5	Local AI Infrastructure & Agentic Workflows	Week 9
S6	Build Day, Governance & Programme Roadmap	Week 11

04 — SESSION DETAIL

Session Details

SESSION 01

Foundations Audit, Strategic Framing & Prompt Engineering

Week 1

MORNING — AUDIT & STRATEGIC FRAMING

- Map SASA's full data landscape: R7106, LIMS, Autolab, mill reports
- Identify integration targets and highest-value AI use cases
- Configure Claude Pro shared account and workspaces
- Review Microsoft Copilot licensing and overlap with Claude
- Establish consulting log and shared SharePoint structure

PROMPT ENGINEERING — AFTERNOON

Foundations of Effective Prompting

- RCTF anatomy: Role, Context, Task, Format — the four levers of every prompt
- XML tags for structured outputs and multi-step tasks
- Understanding Claude's 200K context window and what fits inside it
- Iteration workflow: draft → test → refine → save
- Common failure modes and how to diagnose them
- Establish shared SharePoint prompt library (seed with 10–15 entries)

SESSION 1 DELIVERABLE

- ✓ SASA data & systems map with integration targets annotated
- ✓ Consulting log (shared, living document)
- ✓ Seed prompt library — 10–15 tested entries in SharePoint

SESSION 02

Excel, Reporting & AI-Generated Workflows

Week 3

MORNING — EXCEL CO-PILOT & PROMPT LIBRARY

- Rebuild one live SASA Excel report using Claude as co-pilot
- Copilot Excel comparison — where each tool excels
- Build the exception-flagging template (out-of-spec values, missing data)

PROMPT ENGINEERING — EMBEDDED IN MORNING

Parameterised Reporting Prompts

- Parameterised templates: swap mill, week, or metric without rewriting
- Excel prompt library: summarise / flag / extract / compare — four core patterns
- Version discipline: naming, dating, and archiving prompt iterations

AFTERNOON — NARRATIVE REPORTING & DIRECTOR REVIEW

- Audit the full reporting burden — ROI conversation with management
- Narrative workflow: data → prompt → draft → director sign-off
- Copilot PowerPoint integration for report presentation
- Build report commentary template in SASA house style
- Produce one complete AI-assisted report draft, reviewed by director in-session

SESSION 2 DELIVERABLE

- ✓ AI-augmented Excel report (one live SASA report, rebuilt)
- ✓ Exception-flagging template
- ✓ Excel prompt library (4 core patterns, versioned)
- ✓ Report commentary template in SASA house style
- ✓ One complete AI-assisted report draft, director-reviewed in session

SESSION 03

Working with Large Datasets, RAG & Context Management

Week 5

MORNING — RAG INTRODUCTION & DOCUMENT INDEXING

- Introduce RAG (Retrieval-Augmented Generation) — AI as a librarian for SASA's archive
- Work with SASA's multi-year dataset — R7106, Autolab, LIMS exports
- Build a working document index: chunk, embed, retrieve
- Live demonstration: query 10 years of data without uploading the full dataset

PROMPT ENGINEERING — AFTERNOON

Context Window Management & Multi-Document Prompting

- Context window choices: paste vs. chunk vs. RAG — when to use each
- Chunking strategies for large SASA reports and multi-centre data
- Summarisation chains: reducing 50-page reports to actionable briefs
- Synthesis vs. extraction — knowing which prompt pattern to apply
- Multi-document prompting: cross-referencing mill reports and LIMS output
- Build the SASA data-preparation checklist (LIMS, Autolab, mill formats)

High-value session for online supervisors — participants working with large datasets across multiple mill sites will find this session particularly applicable to day-to-day analytical work.

SESSION 3 DELIVERABLE

- ✓ SASA data-preparation checklist (LIMS, Autolab, mill formats)
- ✓ Working RAG demonstration on SASA historical data
- ✓ Context management reference card

SESSION 04

Python & Automation Pipelines

Week 7

WHAT WE BUILD IN THIS SESSION

- Consulting frame: managers direct, AI writes — Python as the execution layer
- Build an AI-generated Python pipeline: LIMS export → transformation → flagged output
- Introduction to Node.js for API connectors and SharePoint integration
- Version control basics: saving, sharing, and rolling back AI-generated code
- Compile automation candidate list per team area — direct input to Session 6 build day

PRE-SESSION REQUIREMENT

Python, Node.js, and VS Code must be pre-installed on all participant machines before this session. Richard will provide installation instructions two weeks in advance.

SESSION 4 DELIVERABLE

✓ Working LIMS pipeline stub (LIMS export → transformation → flagged output)

✓ Automation candidate list per team area (input to Session 6 build day)

SESSION 05

Local AI Infrastructure & Agentic Workflows

Week 9

MORNING — LOCAL AI INFRASTRUCTURE

- Review IT meeting findings from pre-session infrastructure check
- Install Ollama and Open WebUI on SASA server
- Browser-based access for all participants — no software on individual machines
- Live demo: local RAG query on internal SASA documents — no internet required
- Establish SASA data governance framework: cloud vs. on-premise policy

AGENTIC WORKFLOWS — AFTERNOON

From Prompting to Acting — SASA's First Agent Pipeline

- What is an agent? The ask vs. act distinction — AI that takes steps, not just answers questions
- Introduction to n8n: open-source, self-hosted, zero per-seat cost
- Build a SASA agent pipeline: data input → exception check → draft alert → SharePoint / file output
- Connect to the Session 4 Python output — end-to-end pipeline demonstration
- Design SASA's first agentic use case specification: inputs / triggers / outputs / human review gate
- Server specification notice: minimum server requirements for production Ollama + n8n deployment

SESSION 5 DELIVERABLE

- ✓ Local AI server operational (Ollama + Open WebUI)
- ✓ Working n8n agent pipeline demonstration
- ✓ Agentic use case specification document
- ✓ SASA data governance policy (cloud vs. on-premise classification)

SESSION 06 — CLOSE

Build Day, Governance & Programme Roadmap

Week 11

MORNING — BUILD DAY

- Teams self-select into 2–3 build groups from the Session 4 automation candidate lists
- Build targets: Excel exception report / Python pipeline / local RAG / n8n agent
- Richard rotates as senior consultant — every group gets dedicated support
- Advanced participants lead their groups — internal capability, not consultant dependency
- Goal: every group ships something usable from Monday

AFTERNOON — GOVERNANCE & ROADMAP

- Showcase: 5 minutes per build group — demo, not slides
- Review Session 1 targets: what we set out to do, what we delivered
- Governance framework: acceptable use / data classification / prompt ownership
- Operating model: who owns what after the programme ends
- Skills continuity plan: how to keep building without the consultant
- 12-month integration roadmap: 3–5 targets with owners and review dates
- Individual 90-day action plans for every participant

SESSION 6 DELIVERABLE

✓ Team-built working tools (one per build group, ready Monday)
✓ SASA AI data governance policy (finalised)
✓ 12-month integration roadmap with owners
✓ Individual 90-day action plans

05 — SUMMARY

Programme Deliverables

SESSION	DELIVERABLE
S1	SASA data & systems map · Consulting log · Seed prompt library (10–15 entries)
S2	AI-augmented Excel report · Exception-flagging template · Excel prompt library · Report commentary template · Director-reviewed report draft
S3	Data-preparation checklist · Working RAG demonstration · Context management reference card
S4	Working LIMS pipeline stub · Automation candidate list per team area
S5	Local AI server (Ollama + Open WebUI) · Working n8n agent pipeline · Agentic use case specification · Data governance policy
S6	Team-built working tools · Data governance policy (finalised) · 12-month roadmap · Individual 90-day action plans

06 — OPTIONAL EXTENSION

Extension Programme — Sessions 7–10

OPTIONAL — AVAILABLE ON COMPLETION OF THE 6-SESSION PROGRAMME

Advanced Integration: Sessions 7–10

Four additional sessions at **R3,500 per session** — **R16,100 incl. VAT** for the full extension. Sessions invoiced individually on completion.

- S7 LIMS & Autolab deep integration — connecting live source systems to AI pipelines
- S8 Advanced agentic workflows — scheduling, error handling, and multi-step orchestration
- S9 Source system integration & production deployment — moving from stub to live
- S10 Capstone review & extended roadmap — governance audit and 24-month planning

07 — TECHNICAL

Technical Requirements

AI Licensing

CLAUDE PRO

One shared Claude Pro account is required for the programme. Current cost is approximately **R1,700 per month** (billed in USD, subject to exchange rate). This covers all participants via a shared workspace. No other AI subscription is required for the core six-session programme.

Software (All Open-Source, Zero Per-Seat Cost)

TOOL	PURPOSE	COST	REQUIRED BY
Python 3.11+	Automation pipelines and data transformation	Free	Session 4
Node.js 20 LTS	API connectors and SharePoint integration	Free	Session 4
VS Code	Code editor for all AI-generated code	Free	Session 4
n8n (self-hosted)	Agentic workflow automation and pipeline orchestration	Free (self-hosted)	Session 5
Ollama	Local LLM inference on SASA server	Free	Session 5
Open WebUI	Browser-based interface for local AI (no client install)	Free	Session 5

Server Specification (Sessions 5–6 and beyond)

COMPONENT	MINIMUM	RECOMMENDED
CPU	8-core modern processor	16-core (AMD EPYC or Intel Xeon)
RAM	32 GB	64 GB
Storage	500 GB SSD	2 TB NVMe SSD
GPU (optional)	None (CPU inference)	NVIDIA RTX 4090 or A-series for speed

COMPONENT	MINIMUM	RECOMMENDED
OS	Ubuntu 22.04 LTS	Ubuntu 24.04 LTS
Network	Internal LAN access	Isolated VLAN with firewall policy

08 — INVESTMENT

Investment

ITEM	RATE	SESSIONS	AMOUNT
Applied AI Consulting — 6-session programme	R3,500 per session	6 × 7 hrs	R21,000
VAT @ 15%			R3,150
Total Investment incl. VAT			**R24,150**

Sessions are invoiced individually on completion. No upfront payment required. Extension sessions (S7–S10) are priced at R3,500 each (R16,100 incl. VAT for all four) and are entirely optional.

09 — REFERENCE

Glossary of Terms

For participants new to AI tooling, this glossary defines the key terms used throughout this proposal and the programme.

TERM	DEFINITION
AI	Artificial Intelligence — software systems that perform tasks typically requiring human reasoning, such as reading, summarising, and generating text.
LLM	Large Language Model — the class of AI model underlying tools like Claude and ChatGPT. Trained on vast text corpora; responds to natural language instructions.

TERM	DEFINITION
Claude	Anthropic's large language model. Noted for long context windows, instruction-following accuracy, and safety. Used as the primary AI tool in this programme.
Claude Pro	The paid subscription tier of Claude (~R1,700/month). Provides expanded context, higher usage limits, and shared workspace functionality.
Prompt	The instruction or question given to an AI model. Quality of output is directly determined by quality of the prompt.
Prompt Engineering	The craft of writing structured, effective prompts to get reliable, accurate, and useful responses from an AI model.
Context Window	The maximum amount of text an AI model can read and reason over in a single interaction. Claude's context window is approximately 200,000 tokens (~150,000 words).
RAG	Retrieval-Augmented Generation — a technique where relevant documents are retrieved from a database and inserted into the prompt, enabling AI to answer questions about large document sets without fine-tuning.
Token	The basic unit of text processed by an LLM. Roughly 4 characters or ¾ of a word. Context windows and costs are measured in tokens.
Copilot	Microsoft's AI assistant integrated into Office 365 (Word, Excel, PowerPoint, Teams). Complements Claude for document and spreadsheet workflows.
API	Application Programming Interface — a defined connection point that allows two software systems to communicate. Used to connect Python pipelines to LIMS and SharePoint.
Python	A free, widely-used programming language favoured for data processing and automation. AI can write Python code; managers direct the logic.
Node.js	A JavaScript runtime used for building API connectors and web-based tools. Used in this programme for SharePoint integration.

TERM	DEFINITION
VS Code	Visual Studio Code — a free, industry-standard code editor. Used to view, edit, and run AI-generated Python and Node.js code.
Ollama	A free, open-source tool that runs large language models locally on a server. Enables SASA to use AI on internal documents without sending data to the cloud.
Open WebUI	A browser-based interface for Ollama. Allows all participants to access the local AI model via a web browser — no software installed on individual machines.
Agent / Agentic Workflow	An AI system configured to take a sequence of actions autonomously — not just answer a question, but execute a multi-step task (e.g., check data → draft alert → send to SharePoint).
n8n	An open-source, self-hosted workflow automation tool. Used to build SASA's agent pipelines without per-seat licensing costs. Connects AI outputs to internal systems.
LIMS	Laboratory Information Management System — SASA's system for recording and managing laboratory sample data, including quality and compliance records.
Autolab	SASA's automated laboratory analysis system. Produces structured data outputs that are used as inputs to AI pipelines in this programme.

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